PCB and wiring harness

The rig has no loose wires. One **main controller PCB** sits on the plant room's back wall. Every sensor and actuator connects to it through its own **flat FR4 strip** with copper tracks, as drawn in the 3D model and the STEP file (*PCB wiring harness* sub-assembly, `PCB strip <tag>` parts). This page gives the connector pin-out of every strip, the main-board layout rules and the build order.

The schematics are `03-wiring-power.svg` and `03-wiring-controller.svg`. The pin map is `02-io-list.csv`, and it matches `../firmware/include/config.h`.

Main PCB

- 2 layers, FR4 1.6 mm, 1 oz copper, about 180 × 120 mm, mounted on the back wall at the PCB position in the CAD (`PCB = {x −300 … −120 mm, y 270 … 390 mm}` from the box centre).
- **Bottom layer: solid 0 V plane.** The 24 V bus runs on the top layer at ≥ 2.5 mm width. The ACS712 modules sit right next to their fuses.
- Keep the ADC inputs (GPIO1, 2, 4–7) away from the fan PWM and heater PWM tracks. Put a 100 nF capacitor at each ADC pin and a 10 µF capacitor on each ACS712 output.
- Track widths (IPC-2221, 1 oz, 10 °C rise): signals 0.25 mm; fans 0.4 mm; PTC and compressor ≥ 1.0 mm; 24 V bus ≥ 2.5 mm (or add solder over the track).
- Mount the ESP32-S3-DevKitC-1 on 2 × 22-pin female headers so it can be swapped. Its USB-C port must be reachable from outside the box for flashing and telemetry.

Connectors and strips

Connectors are JST-XH (2.5 mm pitch) unless noted. Pin 1 is marked on the board.

J	Strip (CAD part)	Pins	Signals (pin 1 → n)	Current	Track width on the strip
J1	PSU input	2 (screw terminal)	+24 V, 0 V	6.5 A	wire 1.0 mm²
J2	PCB strip: FAN-101	4	+24 V, 0 V, PWM (open drain), TACH	0.6 A	1.0 / 1.0 / 0.25 / 0.25 mm
J3	PCB strip: FAN-102	4	+24 V, 0 V, PWM, TACH	0.4 A	1.0 / 1.0 / 0.25 / 0.25
J4	PCB strip: FAN-103 + FAN-103B	4	+24 V (through DV-101's SPDT end switch: NO → FAN-103, NC → FAN-103B), 0 V, PWM (both fans), TACH (both, wired-OR)	0.4 A	1.0 / 1.0 / 0.25 / 0.25
J5	PCB strip: DV-101	2	+24 V, SW− (MOSFET, SS34 flyback)	0.2 A	0.5 / 0.5
J6	PCB strip: CMP-101	3	+24 V (through CT-101), 0 V, SPEED 0–5 V	1.1 A	1.0 / 1.0 / 0.25
J7	PCB strip: RV-101 *(on the CMP strip)*	2	+24 V, SW− (MOSFET)	0.2 A	0.5 / 0.5

J	Strip (CAD part)	Pins	Signals (pin 1 → n)	Current	Track width on the strip
J8	PCB strip: EEV-101	4	A1, A2, B1, B2 (DRV8825)	0.3 A	0.5 each
J9	PCB strip: HTR-101	2	+24 V (through CT-103), SW−	1.3 A	1.0 / 1.0
J10	PCB strip: HUM-101	2	+24 V (through CT-103), SW−	0.95 A	1.0 / 1.0
J11	PCB strip: FT-101	3	+24 V, 0 V, Vout	15 mA	0.25 each
J12	PCB strip: FT-102	3	+24 V, 0 V, Vout	15 mA	0.25 each
J13	PCB strips: TT-109, TT-103, TT-104 (daisy-chained)	3	+3.3 V, 0 V, DQ (bus A)	< 5 mA	0.25 each
J14	PCB strips: TT-107, TT-108	3	+3.3 V, 0 V, DQ (bus B)	< 5 mA	0.25 each
J15	PCB strip: TT-101 (MT-101)	4	+3.3 V, 0 V, SDA0, SCL0	< 2 mA	0.25 each
J16	PCB strip: TT-105	4	+3.3 V, 0 V, SDA1, SCL1	< 2 mA	0.25
J17	PCB strip: PDT-101	4	+3.3 V, 0 V, SDA2, SCL2	< 5 mA	0.25
J18	PCB strip: PDT-102	4	+3.3 V, 0 V, SDA3, SCL3	< 5 mA	0.25
J19	PCB strip: TT-106	4	+3.3 V, 0 V, SDA4, SCL4	< 2 mA	0.25
J20	PCB strip: PT-101	7	+5 V, 0 V, SDA5, SCL5, SDA6, SCL6, SEL (0 V)	0.2 A	0.5 / 0.5 / 0.25 …
J24	PCB strip: PT-102	5	+5 V, 0 V, SDA7, SCL7, SEL (0 V)	0.1 A	0.5 / 0.5 / 0.25 …
J21	PCB strip: HMI-101	4	+5 V, 0 V, TX (GPIO17), RX (GPIO18 via 1 k)	0.5 A	0.5 / 0.5 / 0.25 / 0.25
J22	PCB strip: CT (on board)	—	the current sensors are on the main PCB	—	—
J23	PCB strip: PSU	—	the 24 V feed from the PSU to J1	6.5 A	wire 1.0 mm²

There are **no strips or connectors** for the R32 receiver, suction accumulator, filter-drier, sight glass or charging ports, nor for the BIBO HEPA housings, the LAF hood or the Cu plate core. They are mechanical refrigerant parts. The pass box and the vial filling machine are stand-alone and are not wired to the rig either.

Strip construction

- 1.0 mm FR4 (flex-rigid where a strip bends around a corner), copper tracks on the top side, solder mask, tinned pads at both ends.
- Track spacing ≥ 0.5 mm for 24 V. I²C pairs run side by side with 0 V on both sides. Keep each I²C strip under 0.5 m; that is why every device has its own mux channel.
- The device end of each strip is soldered to the sensor breakout or terminated in a small JST. The board end is a JST-XH plug.
- Fix the strips to the acrylic with 3M VHB at about 100 mm spacing, along the routes shown in the 3D model (floor runs, then risers to each device).

Build order

1. Assemble the main PCB and fit the fuses, but don't plug the ESP32 in yet. Power it at 24 V through a current-limited bench supply (0.3 A). Check 5 V on the buck output and that every fuse is intact.
2. Plug in the ESP32-S3, flash the firmware (`firmware/README.md`), and open the monitor. Connect the I²C strips one at a time: the boot **SCAN** must show each device on its channel.
3. Connect the 1-Wire probes and copy the printed ROM codes into `config.h`. Connect the D6F sensors and check that 0.5 V reads as zero flow.
4. Connect the fans (J2–J4), then DV-101 (J5). With `RUN on` and `MAN L1 30`, check the tach and the flow readings. `MAN L3 0` must drive DV-101 to the air pump and start FAN-103; `MAN L3 100` (or pulling J5) must let it spring to the HEPA bypass and start FAN-103B.
5. The compressor (J6), valve (J7) and EEV (J8) come last, after the refrigerant circuit has been leak-tested and charged (`08-build-and-commissioning.md`).